

LIVING MATTER IS THE MOST POWERFUL GEOLOGICAL FORCE

THE BIOSPHERE

IN CONTEXT

KEY FIGURE

Vladimir Vernadsky (1863–1945)

BEFORE

1785 Scottish geologist James Hutton proposes that in order to understand Earth, all of its interactions should be studied.

1875 Austrian geologist Eduard Suess first uses the term “biosphere” to describe “the place on Earth’s surface where life dwells.”

AFTER

1928 In *Methodology of Systematics*, Russian zoologist Vladimir Beklemishe warns that humanity’s future is irrevocably linked to the preservation of the biosphere.

1974 British scientist James Lovelock and American biologist Lynn Margulis first publish their Gaia hypothesis—the idea of Earth as a living entity.

Earth has four interacting subsystems: the lithosphere, Earth’s rigid, rocky outer shell; the hydrosphere, which comprises all water on the planet’s surface; the atmosphere, formed by layers of surrounding gases; and the